Crop Yield Optimizer Report

Paddy | BPT 5204 (Sona Masuri)

Plan Summary

Farmer ID:	farmer_001
Plan ID:	yopt_0b009392
Crop:	Paddy
Variety:	BPT 5204 (Sona Masuri)
Area:	2.5 acres
Stage:	planning
Confidence:	75% (self-assessed)

Advisory report. All recommendations must be validated with local KVK officer before application.

Contents:

1. Variety Selection & Land Preparation

BPT 5204 is the strongest choice for this 2.5-acre kharif paddy plot in Telangana: it commands a consistent market premium of Rs.2,200-2,800/quintal at Hyderabad Monda Market and Suryapet mandi, making it the most bankable grain type for a smallholder. Its 130-145 day duration fits the kharif window cleanly, and with a single bore well on a black cotton vertisol, the AWD (Alternate Wetting-Drying) regime will stretch water availability without the labour overhead of full SRI - all within the Rs.60,000 investment cap. While BPT 5204 requires careful BPH monitoring under heavy nitrogen, disciplined 4-split fertilizer scheduling and pheromone-trap-based IPM (total ~Rs.4,000 incremental) will hold yield at 4.5-5 t/acre, generating Rs.55,000-62,000 gross per acre cycle.

Land Preparation

Soil test needed:	Yes - obtain results before transplanting
Soil amendments:	Basal FYM 2 tonnes/acre (well-decomposed cattle manure) - critical on black cotton to improve aggregate structure and drainage, DAP 50 kg/acre at transplanting (basal P + 25% N), MOP 35 kg/acre at basal (black cotton can have adequate native K - confirm via soil test before applying full dose), ZnSO4 10 kg/acre broadcast pre-puddling - Zn deficiency is near-universal on Telangana vertisols, Neem cake 40 kg/acre broadcast at puddling - suppresses root nematodes and slow-releases N
Tillage / bed type:	Raised-bed / bunded plot with 30 cm raised peripheral bunds - black cotton vertisol has heavy clay that waterlogging-stresses roots in monsoon peaks; raised bunds allow faster drainage of excess standing water while AWD tubes regulate internal moisture
Drainage:	Install at least 2 perforated drainage / AWD monitoring tubes (PVC pipe, 20 cm long, perforated below 15 cm) per 0.5-acre block before transplanting. These allow the farmer to observe when water drops to 15 cm below soil surface before re-irrigating - essential for AWD regime with a single bore well. Cut a shallow drainage outlet channel (25 cm wide × 20 cm deep) on the low end of each plot for emergency flood drainage during heavy monsoon events. Pre-monsoon deep plowing (2× cross passes) in May will break the subsoil clay pan, reduce water-holding lag, and improve percolation.

2. Spacing, Water & Irrigation

Plant spacing:	0.25 m (row) x 0.25 m (plant)
Plant density:	64,000 plants / acre
Geometry notes:	SRI protocol: single seedling per hill, 25×25 cm square grid. Transplant at 8-12 days old (2-leaf stage). Black cotton vertisol - raised peripheral bunds (30 cm) to manage waterlogging peaks. AWD monitoring tubes (PVC, perforated below 15 cm) installed at 1 per 0.5-acre block before transplanting. Square grid allows cono-weeder / rotary weeder passes in both directions - critical for weed management in SRI.

Water Regime

Method: AWD

Single bore well on 2.5 acres of black cotton vertisol - continuous flood is not viable (water table drawdown risk) and vertisol heavy clay creates waterlogging stress under standing flood. AWD paired with SRI spacing delivers 25-30% water saving vs continuous flood with zero yield penalty and often a 10-15% yield uplift on BPT 5204. Perforated PVC monitoring tubes allow the farmer to observe when water drops to 15 cm below soil surface before re-irrigating - essential discipline with a single bore well. Re-flood only when tube reads dry at 15 cm depth. Maintain shallow flood (3-5 cm) only from panicle initiation through grain fill, then drain 7-10 days pre-harvest.

Water savings vs flood:	28%
Yield impact:	+12%
Setup investment:	Rs.2,500-Rs.4,000 for PVC AWD tubes (4 tubes × 2.5 acres), drainage channel digging (manual, NREGA-eligible). Drip not required for paddy AWD - gravity-fed

bore well with bunded plots suffices.

Monthly Irrigation Schedule

Month	Depth (mm)	Frequency / Notes
May	0.0	No irrigation - pre-monsoon deep plowing (2× cross passes). Solarize field. Install AWD tubes and drainage outlet channels.
June	50.0	Pre-transplant puddling only (2 passes). Flood nursery beds daily for seedling establishment. Bore well: 1 full pump cycle (~3 hours) for initial field saturation before transplanting at month end.
July	40.0	AWD cycle begins post-transplanting. Allow field to drain after each 5 cm flood until tube reads 15 cm below surface (~every 5-7 days depending on monsoon). Re-irrigate only when tube reads dry at 15 cm. Expect 3-4 pump cycles/month. Monsoon rain replaces most irrigation - monitor closely.
August	35.0	Continue AWD. Active tillering to panicle initiation phase. Weed with cono-weeder every 10-12 days for first 35 DAT. Bore well supplementation only on dry spells >5 days - 1-2 pump cycles/month typical in Suryapet kharif.
September	60.0	Shift to shallow continuous flood (3-5 cm standing water) from panicle initiation (DAT 50-60) through grain fill. This is the critical yield-protection window - do NOT allow field to dry out. Bore well: daily or every-other-day pumping if monsoon weakens (late-September break common in Suryapet).
October	45.0	Maintain 3-5 cm shallow flood through booting and 50% grain fill (roughly DAT 70-90). Begin gradual drainage drawdown after 80% grain fill. Resume AWD-style management in last 2 weeks. Target field drain-down by DAT 120-125.
November	0.0	Final drainage: drain field completely 7-10 days before harvest (target: DAT 128-135 for BPT 5204, 130-145 day variety). No irrigation. Combine access requires firm soil. Harvest at 20-22% grain moisture.

Growth-Stage Overrides

- PANICLE INITIATION (DAT 50-60): Switch from AWD to shallow continuous flood (3-5 cm). This is non-negotiable - water stress at PI destroys spikelet number irreversibly. Check AWD tube: if not flooded, pump immediately.
- BOOTING (DAT 70-80): Maintain shallow flood. Do NOT allow drying - bud desiccation causes empty grains (white heads).
- GRAIN FILL (DAT 80-100): Maintain 3-5 cm flood. Foliar KH_2PO_4 1% if leaf yellowing appears - prevents late K deficiency common on continuous-paddy vertisols.
- MONSOON BREAK (late September): If SW monsoon withdraws early (common in Suryapet), switch bore well to daily irrigation to maintain flood at PI/booting stages. This is the highest-risk window for a single-bore-well farm.
- PRE-HARVEST DRAIN (DAT 120-125): Drain completely. Do not re-irrigate. Firm soil prevents combine sinkage and reduces threshing loss from over-moist grain.

3. Nutrition Program

Stage	DAP Range	Fertilizers Applied
Pre-Puddling Basal (Land Prep)	DAP 0 (before puddling, May-June)	FYM 2 tonnes/acre - broadcast and incorporate with first plow pass. Black cotton vertisol amendment for aggregate structure and drainage., $ZnSO_4$ 10 kg/acre - broadcast pre-puddling. Near-universal Zn deficiency on Telangana vertisols; soil application more persistent than foliar alone., Neem cake 40 kg/acre - broadcast at puddling. Suppresses root nematodes, slow-releases N over 60-90 days, improves soil biology.
Transplanting Basal (Inorganic)	DAT 0 (at transplanting, late June - early July)	DAP 50 kg/acre - provides basal P (23 kg P_2O_5) + 25% of total N (9 kg N). Broadcast into standing water 24h before or at transplanting, incorporate lightly., MOP 35 kg/acre - provides K_2O 21 kg. Apply ONLY if soil test K < 250 ppm (black cotton often adequate - verify). Reduces risk of K deficiency at grain fill., Urea 20 kg/acre - completes the basal 25% N fraction (together with DAP N). Broadcast on moist field, not flooded standing water, to reduce volatilization.

Stage	DAP Range	Fertilizers Applied
Active Tillering Top-Dress	DAT 21-25	Urea 45 kg/acre - delivers 35% of total N (approx 21 kg N). This is the single most yield-critical fertilizer operation. Apply only if LCC (Leaf Color Chart) score <= 4., Check LCC on 10 random tillers before applying - if score > 4, delay 5 days and re-check. Saves 15-20 kg urea/acre on average and reduces BPH attraction.
Panicle Initiation Top-Dress	DAT 50-60	Urea 30 kg/acre - delivers 25% of total N (~14 kg N). Apply at panicle initiation stage (flag leaf visible)., ZnSO4 0.5% foliar spray - 500 g ZnSO4 dissolved in 100 L water, spray 2 knapsacks/acre. Applied at PI, corrects mid-season Zn deficiency that causes 'khaira' (interveinal chlorosis) and reduces spikelet number by 10-15%., KNO3 1% foliar (optional, if leaf tips showing K stress) - 1 kg KNO3 in 100 L water. Early K correction at PI prevents late-season lodging on black cotton.
Booting / Late Top-Dress	DAT 70-80	Urea 15 kg/acre - delivers final 15% of total N (~7 kg N). Apply ONLY if LCC score <= 4 on flag leaf. Skip entirely if LCC > 4 - excess late N increases BPH attraction and blast risk., KH2PO4 1% foliar (if interveinal yellowing visible) - 1 kg KH2PO4 in 100 L water. Corrects late-season K+P deficiency; improves grain weight and reduces hollow-grain syndrome common in BPT 5204 on vertisols.
Grain Fill Foliar Correction	DAT 85-100	KH2PO4 1% foliar - 1 kg in 100 L water, spray if yellowing or flag leaf pale. Improves 1,000-grain weight., Silicon (potassium silicate 0.2% OR rice husk ash slurry) - optional but recommended on black cotton BPT 5204. Strengthens culm walls, reduces lodging risk and stem borer entry. Apply once at DAT 90.
IPM Monitoring Programme (Season-long)	DAT 25 - harvest	Pheromone traps for BPH (Nilaparvata lugens): 4 traps per 2.5 acres. Check twice weekly from DAT 25. Action threshold: >=10 BPH per hill (or 5/hill if natural enemies absent). Do NOT spray pyrethroids - causes BPH resurgence., Yellow stem borer (Scirpophaga incertulas): Trichogramma japonicum release at 50,000/acre at DAT 25, 32, 39 (3 weekly releases). Supplement with pheromone traps (5/acre) - action at >5 moths/trap/week: cartap hydrochloride 4% GR at 10 kg/acre into standing water., Blast (Pyricularia): Scout for diamond-shaped gray-center lesions weekly from DAT 40. If detected: tricyclazole 75% WP at 120 g/acre immediately. BPT 5204 is susceptible - do not delay., Sheath blight (Rhizoctonia): Check lower sheaths from DAT 30. If spreading upward: validamycin 3% L at 400 ml/acre. Cultural: maintain SRI wide spacing (reduces canopy humidity vs conventional dense planting)., False smut prevention (Ustilaginoidea): propiconazole 25% EC at 200 ml/acre at booting (DAT 72) and 50% heading (DAT 85) - increasingly common on Telangana vertisols with high N + humidity.

- Pre-Puddling Basal (Land Prep): Complete all basal organic + micronutrient applications BEFORE final puddling pass so amendments are incorporated into the anaerobic seed-bed layer. Soil test result should guide MOP decision - black cotton can have high native K (>250 ppm); if confirmed, reduce or skip MOP entirely and redirect budget to neem cake or ZnSO4.
- Transplanting Basal (Inorganic): Total basal N ~25% of 120 kg/acre target = 30 kg N. DAP contributes ~9 kg N + 23 kg P2O5. Urea 20 kg = ~9 kg N. Neem cake (pre-puddling) releases remaining ~12 kg N slowly. Black cotton: drain field 24h before urea application, then re-flood 24h after - reduces NH3 volatilization significantly.
- Active Tillering Top-Dress: Drain field to moist (no standing water) 24h before application. Broadcast urea evenly. Re-flood to AWD level 24h after application. Do NOT apply into deep standing water - up to 60% N lost to volatilization. This split is the highest-ROI single operation in the entire paddy programme.
- Panicle Initiation Top-Dress: This split drives spikelet number - the primary yield component for BPT 5204. Do not skip or delay. If bore well pumping is constrained during late-September monsoon break, prioritise irrigation over all other operations at this stage.
- Booting / Late Top-Dress: Conservative application - the LCC gate is mandatory before applying. On black cotton vertisols with neem cake + FYM base, N mineralisation is active through August-September; late-N demand is often lower than on red soils. Skipping this split (if LCC > 4) saves Rs.600-800 and actively reduces BPH and blast risk.

- Grain Fill Foliar Correction: No solid fertilizer at this stage. Foliar corrections only. Maintain 3-5 cm standing water through this phase. Harvest prep begins at DAT 120 - final drainage.
- IPM Monitoring Programme (Season-long): IPM is the second-highest ROI intervention after N splits. Pheromone trap network (Rs.2,000 total for 2.5 acres) prevents a single BPH or stem-borer event that can destroy 30-50% of yield. Budget Rs.4,000-5,000 for the full season IPM programme. BPT 5204 is susceptible to both blast and BPH - scouting discipline is non-negotiable.

Nitrogen Split Protocol - 120.0 kg N / acre total

Growth Stage	% of Total N	Day After Transplant (DAP)
basal	25.0%	DAP 0
tillering	35.0%	DAP 22
panicle_initiation	25.0%	DAP 55
booting	15.0%	DAP 75

Foliar Corrections

- $ZnSO_4$ 0.5% foliar at DAT 55 (panicle initiation) - corrects near-universal Zn deficiency on Telangana vertisols; apply as 2 knapsack sprays/acre (100 L total).
- KH_2PO_4 1% foliar at DAT 80-90 (grain fill) - corrects late-season K+P deficiency common on continuous-paddy black cotton; improves 1,000-grain weight for BPT 5204.
- Urea 2% foliar at DAT 22-25 (tillering) - supplemental foliar top-up if LCC score is borderline (3-4); apply only if solid urea broadcast not feasible due to standing water depth.
- ALL N top-dresses gated by LCC (Leaf Color Chart): apply solid urea ONLY if leaf score <= 4. At booting (DAT 75), skip 15% split entirely if LCC > 4 - saves 15-20 kg urea/acre and actively reduces BPH and blast susceptibility on BPT 5204.

4. IPM & Crop Protection

Monthly Pest Calendar

Month	Pest / Disease	Threshold	Organic First	Chemical (last resort)
January	Nil (off-season / land prep)	N/A	Deep summer plowing to expose overwintering pupae and pathogens. Remove crop residues and burn or compost.	
February	Nil (off-season)	N/A	Soil health maintenance: apply well-decomposed FYM (2 tons/acre). Check bore well water availability. Plan seed procurement (BPT 5204 or JGL 24423 for BPH-prone black cotton plots).	
March	Nil (off-season / land prep)	N/A	Cross-plowing 2-3 times. Solarize field edges with clear plastic (15 days) to reduce blast spore bank in soil. Apply neem cake 50 kg/acre during last plowing to suppress soilborne pathogens.	
April	Nil (pre-season)	N/A	Procure certified seed (BPT 5204 preferred for premium mandi; JGL 24423 if BPH history). Order pheromone lures (stem borer, leaf folder). Install field water tube for AWD monitoring.	
May	Termites / soil pests (land preparation)	Any visible termite mounds within field boundary	Mix neem cake (50 kg/acre) into top 15 cm during final plowing. FYM incorporation also enriches beneficial microbes that suppress termite colonies.	Chlorpyrifos 20% EC at 2.5 L/acre mixed with sand, broadcast and plowed in during last tillage pass if termite infestation confirmed.
June	Nursery damping-off (Pythium / Rhizoctonia); Leaf folder (early)	Any damping-off patch > 5% nursery area; leaf folder: > 1 affected leaf/plant in nursery	Seed treatment: soak seeds 24 h in water, drain, treat with Trichoderma viride (4 g/kg seed) + Pseudomonas fluorescens (10 g/kg seed) before sowing. Use raised nursery beds (10-12 cm high) on black cotton soil to prevent waterlogging-induced damping-off. Spray NSKE (Neem Seed Kernel Extract) 5% on nursery leaves for leaf folder control.	Nursery drench with Carbendazim 50% WP at 1 g/L if damping-off spreads beyond 10% area. For leaf folder in nursery: Chlorantraniliprole 18.5% SC at 0.4 ml/L (one spray, targeted).

Month	Pest / Disease	Threshold	Organic First	Chemical (last resort)
July	Brown Planthopper (BPH); Yellow Stem Borer; Leaf Folder; Blast (leaf phase)	BPH: >=10 nymphs+adults per hill (tap 10 hills/acre); Stem borer: >5 moths/trap/week or dead hearts visible; Leaf folder: >10% leaves folded/rolled in field; Blast: any diamond-shaped gray-center lesion on leaves	Place 5 yellow sticky traps/acre at crop canopy height for BPH monitoring. Install 5 pheromone traps/acre (stem borer-specific lures) at transplanting. Release Trichogramma japonicum egg parasitoid at 50,000/acre at 10 and 25 DAT (days after transplanting) for stem borer. Maintain AWD regime (drain to 15 cm below soil between irrigations after active tillering) - reduces BPH habitat and methane. Apply LCC (Leaf Color Chart) check before any N top-dress to avoid BPH-attracting excess foliar N. Do NOT spray synthetic pyrethroids - kills BPH natural predators (spiders, mirid bugs) and causes resurgence.	BPH: Imidacloprid 17.8% SL at 60 ml/acre OR Pymetrozine 50% WG at 120 g/acre (rotate chemistries). Stem borer: Cartap hydrochloride 4% GR at 10 kg/acre into 5 cm standing water. Blast: Tricyclazole 75% WP at 120 g/acre at FIRST symptom - do not delay.
August	BPH (peak pressure); Sheath Blight (Rhizoctonia solani); Stem Borer (active tillering); Blast (neck phase risk begins)	BPH: >=10/hill; Sheath blight: any lesion climbing above lowest leaf sheath; Stem borer: dead heart count > 5% tillers; Neck blast watch: blast history on farm or nearby	Continue BPH scouting twice weekly. Continue AWD - do NOT flood continuously. Apply LCC before the 35% N top-dress split (21-25 DAT). Alley-cut (3 ft gap every 12 ft row) to improve air circulation - reduces sheath blight. Spray foliar ZnSO4 0.5% at panicle initiation (50-60 DAT) - corrects near-universal Zn deficiency in Telangana vertisols, improves spikelet number.	Sheath blight: Validamycin 3% L at 400 ml/acre at first lesion sighting; repeat 15 days later. BPH: Dinotefuran 20% SG at 80 g/acre if imidacloprid was used in July (chemistry rotation mandatory). Stem borer (late whorl): Chlorantraniliprole 0.4% GR at 4 kg/acre into standing water.
September	Neck Blast (Pyricularia oryzae); False Smut (Ustilaginoidea virens); BPH (grain fill); Bacterial Leaf Blight (Xanthomonas oryzae)	Neck blast: any infected panicle neck = immediate action (zero tolerance at booting/heading); False smut: >1% panicles with smut balls; BPH: >=10/hill; BLB: linear yellow lesions from leaf tips	CRITICAL MONTH - neck blast can destroy entire crop. Scout daily at booting. Apply preventive tricyclazole at boot stage even without visible symptoms if blast was present in prior season or on neighboring farms. For BLB: drain field immediately; reduce N application; copper oxychloride 50% WP at 600 g/acre as bacteriostatic. Avoid mechanical injury (field entry with heavy equipment). Remove and burn any false smut-infected panicles before spores spread.	Neck blast prevention (MANDATORY at boot stage): Tricyclazole 75% WP at 120 g/acre. If second spray needed (high-risk): rotate to Kasugamycin 3% SL at 400 ml/acre 10 days later. False smut: Propiconazole 25% EC at 200 ml/acre at booting and again at 50% flowering. BPH: Pymetrozine 50% WG at 120 g/acre (rotate from August chemistry).

Month	Pest / Disease	Threshold	Organic First	Chemical (last resort)
October	Grain discoloration (Helminthosporium + Alternaria complex); Storage pests (post-harvest); BPH declining	Grain discoloration: > 5% grains visibly discolored at dough stage; BPH: continue monitoring until drain-down	Drain field 7-10 days before harvest (critical for combine access + grain quality on black cotton - these soils hold moisture longer than red soils; start drain-down earlier at 10 days). Target harvest at 20-22% grain moisture. Hand-pick and remove any false smut balls visible on panicles before they shatter. Sun-dry harvested grain to 14% before storage. Clean storage area with neem leaf litter lining.	Grain discoloration (if severe at milky-dough stage): Propiconazole 25% EC at 200 ml/acre - last fungicide window before harvest. Observe label-specified pre-harvest interval (PHI) strictly. Storage: treat stored grain with Deltamethrin 2.5% WP dusting (1 g/kg grain) if weevil/moth infestation detected in storage area.
November	Post-harvest / storage pests (weevils, moths)	Any visible weevil or moth in stored grain within 2 weeks of storage	Store in moisture-proof gunny bags or HDPE bags. Mix neem leaves (dry, 1 kg per 100 kg grain) in storage to repel weevils. Sun-dry to confirmed 14% moisture before sealing. Record yield per acre against variety - BPT 5204 benchmark 4-5 t/acre; investigate if below 3.5 t/acre.	Aluminium phosphide (Celphos) tablets at 1 tablet per 100 kg sealed grain bags if weevil infestation confirmed during storage. CAUTION: strict safety protocol for phosphine fumigation - do not use in enclosed living spaces.
December	Nil (post-harvest / rabi planning)	N/A	Deep plowing post-harvest to expose stem borer pupae, BPH egg masses, and blast spore bodies to desiccation. Incorporate paddy straw into soil (do NOT burn - retains Si which strengthens next crop). Plan for intercrop rotation: greengram (moong) rabi cycle on 1 of 2.5 acres if bore well permits - fixes N, breaks BPH cycle, earns Rs.15,000-25,000/acre. Soil test this month (KVK Suryapet free under Soil Health Card scheme) for pH, N, P, K, Zn status before next kharif.	

5. Harvest & Post-harvest

Maturity indicators: 80-85% of grains on panicle turn golden-yellow (BPT 5204 typical at 130-145 days after transplanting); Grain moisture content at 20-22% - test with portable moisture meter before combine entry; Lower leaves dry and straw-colored; upper canopy still slightly green; Grain hardness test: squeeze a grain - no milky exudate, firm chalky kernel; Field drain complete (7-10 days pre-harvest): soil firm enough for combine/harvester without sinking

Harvest method: Paddy combine harvester (self-propelled, small 4-ft cut): recommended over manual for 2.5 acres to reduce harvest loss from shattering (BPT 5204 moderately susceptible to shattering). Drain field 7-10 days prior. Set combine cylinder speed at 600-700 rpm to minimize grain breakage. Harvest in morning (6-10 AM) when humidity is higher - reduces shattering loss by 2-3%. Manual harvest with sickle acceptable but increases labor cost by Rs.3,000-4,000 and harvest duration by 3-4 days, increasing rain-damage risk in Oct-Nov.

Post-harvest Steps

- Immediately sun-dry harvested grain to 14% moisture (from 20-22% at harvest) - spread on threshing floor or tarpaulin for 2-3 days; turn twice daily
- Winnow / clean grain: remove straw, chaff, and immature grains using aspirator or manual winnowing
- Grade for BPT 5204 Sona Masuri premium: remove chalky, broken, and discolored grains (>5% broken triggers mandi grade downgrade)
- Bag in jute sacks (50 kg) and label variety + harvest date for FPO / procurement aggregator
- Store in cool dry location (moisture <14%, temp <30°C) on raised wooden pallets - do not store on earthen floor
- Apply phosphine fumigation tablets (Celphos 3g/tonne) if storage exceeds 30 days - mandatory to prevent weevil (Sitophilus oryzae) damage
- Register with Telangana Civil Supplies for MSP procurement within 30 days of harvest - carry Aadhaar, Rythu Bandhu passbook, and yield estimate

Storage: Black cotton vertisol farm: post-harvest field drying is critical - black soil retains moisture longer than red soil, increasing fungal risk on harvested grain left in field overnight. Never heap grain on bare soil. Maximum on-farm storage: 60 days with fumigation. Beyond 60 days: FCI warehouse or FPO cold storage preferred. BPT 5204 is susceptible to grain discoloration in humid storage - maintain <14% moisture strictly.

6. Economics & Cash Flow

Year-by-Year Yield Benchmarks

Year	Low	High	Unit	Notes
Y1	18.0	25.0	quintals/acre	Year 1 Kharif - black cotton vertisol, single bore well, AWD regime. Low: conventional flood + basic inputs. High: AWD + 4-split N + IPM. SRI full protocol adds further but labor-intensive in Year 1.
Y2	20.0	27.0	quintals/acre	Year 2 - farmer learns water management on black cotton. Soil health slightly improved with stubble incorporation. IPM scouting now routine.
Y3	22.0	29.0	quintals/acre	Year 3 - micronutrient corrections bedded in (Zn, K). AWD discipline improves. BPH resistance managed via chemistry rotation.
Y4	23.0	30.0	quintals/acre	Year 4 - soil organic matter building with green manure / FYM incorporation between seasons. Yield ceiling approach.
Y5	23.0	30.0	quintals/acre	Year 5 - plateau range under AWD + 4-split. SRI practitioners can reach 32-35 q/acre at this stage.
Y6	22.0	30.0	quintals/acre	Year 6 - soil fatigue risk if continuous paddy-paddy rotation. Introduce rabi pulse (greengram/blackgram) to break cycle and restore N.
Y7	22.0	30.0	quintals/acre	Year 7 - maintained with crop rotation; sheath blight pressure may increase in continuous rice plots; validamycin spray routine.
Y8	21.0	29.0	quintals/acre	Year 8 - silicon depletion in continuous paddy; add potassium silicate foliar or paddy husk ash incorporation to arrest lodging trend.
Y9	21.0	29.0	quintals/acre	Year 9 - stable if rotation maintained; bore well water table stress may start showing on black cotton if over-extraction.
Y10	20.0	28.0	quintals/acre	Year 10 - re-evaluate bore well depth and soil health. Deep plowing + green manure recommended. Consider variety diversification (JGL 24423 or RNR 15048 niche) if market access allows.

Cash Flow Projection

Year	Investment (Rs.)	Revenue (Rs.)	Net (Rs.)	Notes
Y1	55,000	126,500	71,500	2.5 acres × Rs.22,000/acre costs. Revenue: 23 q/acre avg × 2.5 acres × Rs.2,300 MSP = Rs.1,32,250 gross; net after costs Rs.71,500. Rythu Bandhu Rs.25,000 (2.5 acres × Rs.10,000) partially offsets investment.
Y2	50,000	138,000	88,000	Slightly lower investment (no new drip/equipment). Yield improves to 24 q/acre avg. Revenue: 24 × 2.5 × Rs.2,300 = Rs.1,38,000.
Y3	50,000	149,500	99,500	Yield 26 q/acre avg. Revenue: 26 × 2.5 × Rs.2,300 = Rs.1,49,500. Incremental IPM savings from pheromone scouting reduce spray costs.

Year	Investment (Rs.)	Revenue (Rs.)	Net (Rs.)	Notes
Y4	52,000	155,250	103,250	Yield 27 q/acre. Revenue: 27 × 2.5 × Rs.2,300 = Rs.1,55,250. Small spike in investment for bore-well maintenance and soil amendment (ZnSO4 + FYM top-up).
Y5	50,000	155,250	105,250	Yield plateau at 27 q/acre. Consistent returns. Considering FPO membership for milling margin opportunity.
Y6	48,000	149,500	101,500	Yield slight dip to 26 q/acre - rabi greengram rotation introduced; paddy input costs dip slightly. Greengram income (Rs.15,000-20,000 additional) not included here.
Y7	50,000	149,500	99,500	Sheath blight management adds Rs.1,500 spray cost. Yield holds at 26 q/acre.
Y8	53,000	143,750	90,750	Yield 25 q/acre - silicon correction investment (Rs.2,000 extra). Bore well motor service cost assumed Rs.3,000.
Y9	50,000	143,750	93,750	Stable. Assumed MSP increments offset any yield stagnation. Revenue at Rs.2,300 MSP baseline (likely higher by Y9).
Y10	55,000	138,000	83,000	Yield 24 q/acre avg - deep plowing + green manure investment Rs.5,000 extra. Variety re-evaluation year. 10-year cumulative net: ~Rs.9.36 L on 2.5 acres.

Production Cost Breakdown

Cost Item	Annual Cost (Rs.)	Notes
Land preparation (deep plowing × 2, leveling, puddling)	7,000	Tractor hire: 2 passes deep plowing + 1 cross plow + puddling on 2.5 acres. Black cotton vertisol needs 2 passes for adequate pulverization. Rs.2,500-3,000/acre = Rs.7,000 total.
Seed (BPT 5204 certified seed)	2,500	SRI protocol: 2 kg/acre × 2.5 acres = 5 kg. Certified BPT 5204 at Rs.500/kg from KVK or govt seed depot. Treat with Bijamruta before sowing.
Nursery management (SRI/transplanted)	2,000	Nursery bed prep, watering, and weeding for 8-12 day SRI seedlings. Includes plastic trays if SRI mat nursery used. Rs.800/acre × 2.5 = Rs.2,000.

Cost Item	Annual Cost (Rs.)	Notes
Fertilizers (DAP, MOP, Urea - 4-split schedule)	12,000	Basal: DAP 50kg + MOP 35kg/acre. Top-dress: Urea 3 splits × ~35kg/acre each. Total per acre ~Rs.4,800. Plus $ZnSO_4$ 10kg/acre Rs.350. 2.5 acres = Rs.12,000 total.
Micronutrients ($ZnSO_4$ foliar, KH_2PO_4 grain fill)	1,500	$ZnSO_4$ 0.5% foliar at panicle initiation + KH_2PO_4 1% at grain fill. Critical for black cotton vertisol where Zn mobility is low at high pH.
IPM inputs (pheromone traps, biocontrol, targeted chemicals)	5,000	BPH: 5 pheromone trap positions + Imidacloprid/Pymetrozine spray 1-2 rounds. Stem borer: Trichogramma japanicum release × 3. Blast: Tricyclazole preventive at booting. Total Rs.2,000/acre × 2.5 = Rs.5,000.
Irrigation (bore well electricity + pump maintenance)	6,000	AWD regime reduces pumping vs continuous flood. Estimated 18 irrigation events × 3h each on 2.5 acres. Electricity at Rs.6/kWh (non-subsidized) or minimal under Telangana free agri power scheme. Motor service Rs.1,500/year included.
Transplanting labor	6,000	SRI transplanting (single seedling, wide spacing, 8-12 day seedlings) is more labor-intensive than conventional. 8-10 persons × 1 day on 2.5 acres × Rs.300/day = Rs.2,400/acre approx. Total Rs.6,000.
Weeding (manual + cono-weeder for SRI AWD)	5,500	SRI AWD has higher weed pressure than continuous flood. Cono-weeder passes × 3 + 1 manual weed at 10-day intervals for first 35 DAT. Pre-emergent Pendimethalin at Rs.400/acre. Total Rs.5,500.

Cost Item	Annual Cost (Rs.)	Notes
Harvest (combine hire + winnowing + bagging)	7,000	Combine hire: Rs.2,000-2,500/acre × 2.5 = Rs.6,000. Jute sacks + bagging labor Rs.500. Drying tarpaulin amortized Rs.500. Total Rs.7,000.
Transport to mandi / Civil Supplies procurement center	3,500	57-60 quintals (2.5 acres × 23-24 q/acre avg). Tractor-trolley hire to nearest TNCSC center (~15-30 km from Suryapet/Nalgonda). Rs.60-70/quintal transport cost.

Benchmark Comparison

Metric	Value	Unit
Your target	27.0	quintals/acre (kharif, BPT 5204)
District average	20.0	quintals/acre (kharif, BPT 5204)
State best practice	35.0	quintals/acre (kharif, BPT 5204)

District average (Suryapet/Nalgonda) is 18-22 q/acre under conventional flood + single basal N + calendar sprays. This plan targets 25-30 q/acre in Year 1 rising to 27-30 q/acre by Year 3, using AWD + 4-split N + pheromone IPM + ZnSO4 correction - achievable without SRI. State best (35 q/acre) is achieved by full-SRI practitioners with optimal water + fertility management; incorporating the SRI lever (Priority 5) brings this farm within range of state best by Year 3-4. The 27 q/acre target represents a 35% improvement over district average and is the realistic Year 1-2 ceiling for this farm given single bore well constraint and first-season AWD adoption. Revenue gap between district average and target: 7 q/acre × 2.5 acres × Rs.2,300 = Rs.40,250/year additional income - recoverable with Rs.9,350 total incremental investment across all Tier 1 levers.

7. Risk Register & Optimization Levers

Risk Register

Risk	Probability	Impact	Mitigation
Brown Planthopper (BPH) outbreak	high	30-50% yield loss if untreated; BPT 5204 is moderately susceptible. On black cotton with high N buildup, risk amplified.	Pheromone trap network (5 positions, 2.5 acres). Scout twice weekly from active tillering. Apply Pymetrozine 50% WG (120g/acre) or Imidacloprid at threshold >=10 BPH/hill. Strictly avoid synthetic pyrethroids. Use 4-split N (LCC-guided) to avoid N excess that fuels BPH. AWD water regime reduces BPH habitat preference.
Waterlogging on black cotton vertisol during peak monsoon (July-August)	high	Root asphyxiation, tillering suppression, 15-25% yield loss; in severe events, crop failure on low-lying patches. Black cotton holds water for 5-7 days after heavy rain.	Install raised bund drainage channels on all field boundaries before kharif. Create a 30 cm outlet drain to allow excess water exit within 48 hours of heavy rain. Monitor weather forecast - drain proactively 24h before forecast heavy rain events. Avoid SRI transplanting in low-lying black cotton patches; use conventional transplanting there.
Neck blast (Pyricularia oryzae) at booting stage	medium	White sterile panicles - 20-40% yield loss. BPT 5204 is blast-susceptible. October humidity in Telangana creates ideal blast window.	Preventive Tricyclazole 75% WP at 120g/acre at boot leaf stage (70-75 DAT). Do NOT apply irrigation in late evening (reduces overnight leaf wetness). Rotate to Kasugamycin if second spray needed (resistance management). Silicon foliar (potassium silicate 2g/L) at panicle initiation strengthens cell walls and reduces blast entry.
Bore well failure / water scarcity during critical crop stages	medium	Single bore well is a single point of failure. Critical stages - panicle initiation and grain fill - require consistent moisture. Failure during grain fill causes chalkiness and 20-30% yield loss.	Identify nearest neighbor bore well as emergency backup (500m radius). Keep bore well motor serviced annually. AWD regime (vs continuous flood) reduces total pumping demand by 20-25%, preserving water table. Mulch inter-rows during dry spells. Install a 1,000L overhead tank to buffer 1-2 day pump outages.

Risk	Probability	Impact	Mitigation
Market price risk - mandi price falling below MSP, delayed procurement	medium	BPT 5204 mandi price can fall to Rs.1,800-2,000/q in glut years. Telangana Civil Supplies procurement delays can force distress mandi sales. On 2.5 acres at 23 q/acre, a Rs.300/q shortfall = Rs.17,000 income loss.	Register with Telangana Civil Supplies (TNCSC) procurement center before harvest. Maintain dry BPT 5204 grain (14% moisture) for quality eligibility. Join nearest FPO for collective bargaining and direct mill/buyer access. Explore custom milling (Rs.800-1,200/tonne margin) if FPO has hulling unit. Storage: fumigate grain if holding beyond 30 days awaiting better price.
Yellow stem borer at tillering / booting stage	medium	Dead hearts at tillering (15-20% yield loss); white heads at booting (severe - up to 30% loss). Common in Suryapet/Nalgonda belt kharif.	Pheromone traps (5/acre) from transplanting. Trichogramma japonicum egg parasitoid release: 50,000/acre × 3 releases at weekly intervals starting tillering. Chemical threshold: >5 moths/trap/week -> Cartap hydrochloride 4% GR at 10 kg/acre OR Chlorantraniliprole 0.4% GR at 4 kg/acre - apply into standing water. Do NOT use endosulfan (banned).
Soil zinc deficiency (common in black cotton vertisol, high pH inhibits Zn availability)	high	Khaira disease symptoms: brown rusty spots on younger leaves, stunted tillering, 10-20% yield reduction. Near-universal in continuous paddy on Telangana black soils.	Apply $ZnSO_4$ 10 kg/acre at basal (Year 1 mandatory). Foliar $ZnSO_4$ 0.5% at panicle initiation every year. Soil test every 2-3 years (free under Soil Health Card). If soil pH >8.0, shift to chelated Zn (EDTA-Zn) foliar - more mobile at high pH than soil-applied sulfate form.

Optimization Levers (ranked by priority)

#	Lever	+Yield	Investment	Difficulty	Payback
1	4-split nitrogen schedule with LCC (Leaf Color Chart) guidance	+20%	Rs.2,000 (extra fertilizer split labor + LCC cards from KVK - free)	easy	Same season - 20% yield uplift on 2.5 acres at MSP = Rs.26,450 additional revenue for Rs.2,000 investment

#	Lever	+Yield	Investment	Difficulty	Payback
2	AWD (Alternate Wetting and Drying) water regime - replace continuous flood	+10%	Rs.800 (perforated PVC field tube × 4 for AWD monitoring, available at hardware shops)	easy	Same season - saves 20-25% water (extends bore well life), slight yield gain, reduces methane footprint for future carbon credit eligibility
3	Pheromone-based IPM scouting (BPH + stem borer traps) replacing calendar sprays	+15%	Rs.2,000 (10 pheromone traps + 3 Trichogramma releases at Rs.400/release)	easy	Same season - prevents 30-50% yield loss in BPH outbreak year; net spray cost reduction Rs.1,500 vs calendar approach
4	Zinc sulfate foliar correction at panicle initiation ($ZnSO_4$ 0.5%)	+8%	Rs.350 ($ZnSO_4$ 10 kg/acre for soil + foliar spray materials)	easy	Same season - addresses near-universal Zn deficiency on black cotton vertisol; 8% yield uplift on spikelet number and grain fill
5	SRI (System of Rice Intensification) full protocol - young seedlings (8-12 days), wide spacing 25×25cm, single seedling per hill	+30%	Rs.5,000 (cono-weeder rental/purchase Rs.3,000 + extra transplanting labor Rs.2,000 on 2.5 acres)	moderate	Year 1 - 30% yield gain at 2.5 acres = +17 quintals × Rs.2,300 = Rs.39,100 additional. Investment Rs.5,000 = 7.8× ROI
6	FPO membership for MSP procurement + custom milling margin access	+0%	Rs.1,000-2,000 FPO membership fee (one-time)	easy	Rs.800-1,200/tonne milling margin on ~5.75 tonnes/year = Rs.4,600-6,900 additional income with zero yield change; also improves procurement priority queue at Civil Supplies

#	Lever	+Yield	Investment	Difficulty	Payback
7	Kharif-Rabi rotation: introduce greengram (Moong) as rabi short-cycle crop post-paddy	+5%	Rs.6,000-8,000/season for greengram (seed + inputs on 2.5 acres)	easy	Greengram net Rs.15,000-25,000/season additional income. Nitrogen fixation benefit improves paddy yield 5% in following kharif. Breaks continuous paddy disease cycle.

8. Devil's Advocate Critique

Overall confidence: 52%

Biggest yield gap driver: Single bore well with no backup water source on black cotton vertisol: any pump failure, power cut, or water table drawdown during the 10-day critical window around panicle initiation (DAT 50-60) will cause irreversible spikelet sterility and abort the 27 q/acre target regardless of how well every other agronomic lever is executed. This is the one risk in the plan that cannot be corrected after the fact.

Why the target IS realistic

- AWD water regime is well-matched to a single bore well on black cotton vertisol - continuous flood would risk bore well depletion and waterlogging root stress, so the protocol directly addresses the two dominant physical constraints of this parcel. AWD has documented zero yield penalty vs continuous flood and a 10-15% uplift on BPT 5204 under good discipline, which alone narrows the gap to the 27 q/acre target.
- The 4-split nitrogen schedule with LCC guidance is the single highest-leverage agronomic lever available in Telangana paddy - moving from a single basal application to 4 timed splits (basal 25%, tillering 35%, PI 25%, booting 15%) consistently delivers 15-25% yield uplift in on-farm trials at Suryapet KVK, and the 20% uplift claimed in the plan is conservative relative to the upper bound. On black cotton soil with moderate native K, the fertilizer response is typically strong.
- The 27 q/acre target is only 35% above the stated district average of 20 q/acre for kharif BPT 5204. This is not a heroic target - it sits squarely in the 'good management' benchmark band (20-27 q/acre) from the KB, and the KB documents that AWD + 4-split N + IPM routinely achieves 22.5-25 q/acre even without SRI, with best-practice touching 27+ q/acre. The plan's Tier 3 benchmark of 29 q/acre would be exceptional; 27 is achievable.
- 12 pest calendar entries signals a threshold-based IPM calendar rather than calendar-based blanket spraying - this is critical on black cotton vertisol where high native fertility and standing moisture create ideal BPH breeding conditions. Proactive pheromone monitoring + LCC discipline together reduce the risk of the two most common catastrophic yield events (BPH outbreak and lodging from over-fertilization), both of which are especially destructive on BPT 5204 which is classified as moderately susceptible to BPH.
- Black cotton vertisol, despite its waterlogging risk under continuous flood, has high native K and reasonable available P - this means the crop has a strong natural fertility base that responds sharply to targeted N management. With AWD preventing the anaerobic root stress that suppresses nutrient uptake under continuous flood on heavy clay, BPT 5204 on this soil type has consistently shown above-average grain fill when water discipline is maintained.

Why the target might NOT be realistic

- Single bore well is the plan's Achilles heel: black cotton vertisol requires precise AWD discipline - the perforated PVC tube method only works if the farmer re-irrigates promptly when the tube reads dry at 15 cm. A single bore well on 2.5 acres with no backup means any bore well failure (pump burnout, power outage of 3+ days, water table drawdown during a dry spell in August) can strand the crop at panicle initiation or grain fill - the two stages where water stress causes irreversible yield loss. There is no safety net. The plan does not mention a farm pond or alternate water source.
- The 64,000 plants/acre spacing cited in the plan is anomalously high for BPT 5204 under AWD. Standard SRI spacing is 25×25 cm (~64,516 plants/acre in theory) but SRI plants must be transplanted at 8-12 days old as single seedlings - if the farmer is used to conventional transplanting of 25-30 day seedlings in clumps of 3, achieving true SRI plant establishment at this density on black cotton (which puddles poorly and cracks in dry phases) is technically demanding. Failed establishment at even 15-20% of hills would significantly depress tiller count and final yield.
- BPH is flagged as the biggest risk and rightly so - BPT 5204 is moderately BPH-susceptible, and the plan's 4-split N schedule, while optimal for yield, keeps foliar N elevated throughout the season, which is exactly the condition under which BPH populations explode. On black cotton with poor air circulation at dense SRI planting and AWD's moist (not flooded) conditions, BPH nymphs can build from sub-threshold to 15+/hill within 7 days. If the farmer misses even one scouting window (twice-weekly scouting is the requirement but is rarely maintained through monsoon fieldwork pressure), a single BPH flush can erase 25-40% of grain weight.
- Kharif 2025-26 in the Suryapet/Nalgonda corridor has shown erratic onset - if the SW monsoon delivers a dry gap of 15+ days in August (a recurring pattern in Jangaon/Suryapet in recent years), the AWD protocol collapses into unintended drought stress because the single bore well cannot compensate for a multi-week deficit at the active tillering to PI transition. The target of 27 q/acre assumes a normal monsoon season; a 15% rainfall deficit in August reduces BPT 5204 yields by 18-25% in bore-well-dependent systems.

- The plan lists 7 nutrition stages - this is more complex than the standard 4-split KB protocol and adds execution risk. Each additional split requires a field visit, LCC reading, calculation, and timely procurement of inputs. On a 2.5-acre smallholding, the farmer is typically managing multiple parcels and household labor simultaneously. Over-engineering the nutrition schedule increases the probability of missed or mistimed applications, which on BPT 5204 - a variety highly sensitive to late-season N timing - can cause panicle sterility or lodging rather than yield uplift.

9. Week-by-Week Activity Calendar

Derived from N-split protocol, irrigation schedule and IPM calendar. Adjust dates to your actual sowing/transplant date.

Wk	DAP	Growth Stage	Irrigation	Fertilizer / Nutrition	IPM / Scouting
1	0-6	Nursery prep	0.0mm / No irrigation - pre-monso	N 25.0% - basal	Routine scout
2	7-13	Nursery / seedling	0.0mm / No irrigation - pre-monso	-	Nil (off-season / land pr
3	14-20	Seedling growth	0.0mm / No irrigation - pre-monso	-	Routine scout
4	21-27	Transplanting & estab.	0.0mm / No irrigation - pre-monso	N 35.0% - tillering	Routine scout
5	28-34	Transplanting & estab.	50.0mm / Pre-transplant puddling o	-	Routine scout
6	35-41	Active tillering	50.0mm / Pre-transplant puddling o	-	Nil (off-season)
7	42-48	Active tillering	50.0mm / Pre-transplant puddling o	-	Routine scout
8	49-55	Maximum tillering	50.0mm / Pre-transplant puddling o	N 25.0% - panicle_initiation	Routine scout
9	56-62	Maximum tillering	40.0mm / AWD cycle begins post-tra	-	Routine scout
10	63-69	Panicle initiation	40.0mm / AWD cycle begins post-tra	-	Nil (off-season / land pr
11	70-76	Panicle initiation	40.0mm / AWD cycle begins post-tra	N 15.0% - booting	Routine scout
12	77-83	Booting / heading	40.0mm / AWD cycle begins post-tra	-	Routine scout
13	84-90	Booting / heading	35.0mm / Continue AWD. Active till	-	Routine scout
14	91-97	Flowering / grain fill	35.0mm / Continue AWD. Active till	-	Nil (pre-season)
15	98-104	Flowering / grain fill	35.0mm / Continue AWD. Active till	-	Routine scout
16	105-111	Grain maturity / harvest	35.0mm / Continue AWD. Active till	-	Routine scout
17	112-118	Grain maturity / harvest	60.0mm / Shift to shallow continuo	-	Routine scout
18	119-125	Grain maturity / harvest	60.0mm / Shift to shallow continuo	-	Termites / soil pests (la

Key Fertilizer Milestones

- DAP 0 (Week 1): basal - apply 25.0% = 30.0 kg N/acre
- DAP 22 (Week 4): tillering - apply 35.0% = 42.0 kg N/acre
- DAP 55 (Week 8): panicle_initiation - apply 25.0% = 30.0 kg N/acre
- DAP 75 (Week 11): booting - apply 15.0% = 18.0 kg N/acre
- Harvest: 80-85% of grains on panicle turn golden-yellow (BPT 5204 typical at 130-145 days after transplanting); Grain

moisture content at 20-22% - test with portable moisture meter before combine entry; Lower leaves dry and straw-colored; upper canopy still slightly green

Disclaimers & Data Sources

- This plan is advisory only - consult local KVK/ATMA before major investments.
- Prices and yields are estimates based on Telangana averages; actual results vary.
- Govt scheme eligibility subject to current year notifications.
- All chemical recommendations must be validated against current CIB&RC registrations before use.
- Yield projections are estimates based on ICAR/KVK published data. Actual results depend on weather, soil variability, and management quality.
- This report does not constitute a guarantee of income or yields. The farmer bears sole responsibility for all agronomic decisions.